

Tariq Farooq

Senior Data Engineer | Cloud & Big Data Architect | Real-Time Streaming
Data Lakehouse & Governance | ETL/ELT Specialist | HIPAA-Compliant Platforms

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PROFESSIONAL SUMMARY

Senior Data Engineer with **10+ years** of experience architecting enterprise-scale data platforms, real-time streaming pipelines, and **HIPAA-compliant lakehouse systems**. Proven record of processing **2.3M+ records daily** at 99.9% reliability, reducing cloud infrastructure costs by **\$2M+ annually**, and leading zero-downtime migrations at petabyte scale. Expert in **Databricks, Apache Kafka, Spark, Delta Lake, dbt, Airflow, Snowflake, BigQuery**, and multi-cloud architectures across **AWS, Azure, and GCP**. Adept at building federated Data Mesh governance, ML feature stores, and CI/CD-integrated data quality pipelines. Consistent track record of mentoring high-performing engineering teams and translating complex technical solutions into measurable business outcomes.

CORE COMPETENCIES

Data Lakehouse Architecture	Real-Time Streaming Pipelines	HIPAA / PHI Data Compliance
ETL/ELT Pipeline Engineering	Cloud Cost Optimization (FinOps)	Data Mesh & Federated Governance
ML Feature Engineering	Team Mentorship & Leadership	Zero-Downtime Migrations

TECHNICAL SKILLS

- Languages & Scripting:** Python, SQL, Scala, Java, Bash, Shell Scripting, YAML, JSON
- Big Data & Distributed Computing:** Apache Spark, PySpark, Databricks, Apache Flink, Apache Beam, Hadoop (HDFS, Hive), Amazon EMR, Trino, Presto, Delta Lake, Apache Iceberg, Apache Hudi
- Streaming & Event Platforms:** Apache Kafka, Kafka Streams, Debezium, Spark Structured Streaming, AWS Kinesis, Google Pub/Sub, Apache Pulsar, Redpanda, Change Data Capture (CDC)
- ETL/ELT & Orchestration:** Apache Airflow, dbt (data build tool), AWS Glue, Azure Data Factory, Prefect, Apache NiFi, AWS Step Functions, Fivetran, Stitch
- Cloud & Multi-Cloud Platforms:** AWS (S3, EMR, Glue, Lambda, Kinesis, Redshift, Lake Formation, Athena), Azure (Databricks, Synapse Analytics, Event Hubs, ADLS), GCP (BigQuery, Dataflow, Pub/Sub, GCS)
- Lakehouse & Data Storage:** Delta Lake, Apache Iceberg, Apache Hudi, Unity Catalog, Amazon S3, Azure Data Lake Storage (ADLS), Google Cloud Storage, AWS Lake Formation
- Databases & Warehouses:** Snowflake, BigQuery, Amazon Redshift, Azure Synapse Analytics, PostgreSQL, MySQL, SQL Server, MongoDB, Apache Cassandra, DynamoDB, Redis, Elasticsearch
- Data Modeling & Architecture:** Dimensional Modeling, Data Vault 2.0, Star & Snowflake Schemas, CDC Patterns, ERD/UML, Data Mesh, Slowly Changing Dimensions (SCD), One Big Table (OBT)
- DevOps & Infrastructure (IaC):** Terraform, Kubernetes (K8s), Docker, Helm, GitHub Actions, Jenkins, Argo CD, CI/CD Pipelines, Infrastructure as Code (IaC)

- **Observability & Reliability:** Prometheus, Grafana, OpenTelemetry, Datadog, PagerDuty, SLA/SLO Management, Alerting, Incident Response
- **Governance & Data Quality:** Great Expectations, Monte Carlo, DataHub, Apache Atlas, Collibra, Data Lineage, IAM, PHI/PII Protection, HIPAA Compliance, SOC 2, GDPR
- **ML & MLOps:** MLflow, Scikit-learn, TensorFlow, Feature Engineering, Feature Stores, Model Pipelines, Vector Databases (FAISS, Pinecone)
- **BI & Decision Support:** Power BI, Tableau, Looker, Apache Superset, Amazon QuickSight, Metabase, KPI & Executive Dashboards
- **Domain & Compliance:** HIPAA-compliant data platforms, PHI/PII handling, clinical and claims data processing, healthcare interoperability, audit logging, encryption standards, access controls, governance for regulated environments
- **Soft Skills:** Technical Leadership, Cross-functional Collaboration, Stakeholder Communication, Agile / Scrum, System Design, Mentoring & Coaching

TECHNICAL EXPERIENCE

Lead Data Engineer

Truveta, Inc. — *Healthcare data platform company; real-time clinical intelligence for 150+ hospital networks.* May 2022 – Present

- Architected and maintained **HIPAA-compliant lakehouse platforms** on **Databricks** processing **2.3M patient records daily** across 150+ hospital networks, achieving **99.9% uptime** and enabling real-time clinical decision support at scale.
- Reduced cloud infrastructure costs by **\$1.2M annually (40% reduction)** through Spark workload optimization, intelligent storage tiering on **AWS S3**, and auto-scaling compute clusters with SLA-based workload classification.
- Built real-time **CDC pipelines** using **Apache Kafka**, **Debezium**, and **Spark Structured Streaming** to process **500K+ healthcare events per hour** with **sub-5-second latency**, powering live dashboards in Power BI and Tableau.
- Implemented **multi-cloud data architecture** spanning AWS, Azure, and GCP with unified governance via **Unity Catalog** and **AWS Lake Formation**, ensuring consistent PHI/PII controls and audit compliance across **12 petabytes** of healthcare data.
- Established automated **data quality framework** using **Great Expectations** and **Monte Carlo**, reducing data incidents by **73%** through proactive validation of 200+ critical production datasets.
- Designed and deployed **Delta Lake-based feature stores** for ML pipelines, cutting feature engineering time by **60%** and supporting 15+ production ML models via **MLflow** integration.
- Led enterprise **Data Mesh initiative** with federated governance, enabling 8 domain teams to self-serve data products and reducing data onboarding time from 6 weeks to **48 hours**.
- Optimized **Apache Spark jobs** via adaptive query execution, Z-ordering, and partition tuning, improving pipeline performance by **65%** and reducing average query latency.
- Maintained **99.95% SLA adherence** using Prometheus/Grafana monitoring with auto-remediation playbooks resolving **80% of incidents** without manual intervention.
- Mentored **6 data engineers** through code reviews, architecture design sessions, and pair programming, elevating team capability to independently deliver production-grade pipelines.
- Bridged clinical stakeholders and engineering teams by translating complex healthcare workflows into detailed functional specifications and **data mapping documents** for FHIR-based interoperability.

- Directed end-to-end **UAT and validation cycles** for HL7 FHIR resource mapping, ensuring 100% alignment with **CMS ONC certification standards** and customer clinical requirements.

Senior Data Engineer

Starschema — *Global data consultancy; enterprise cloud analytics and data warehouse solutions.*

Jan 2017 – Apr 2022

- Built and scaled **cloud analytics platforms** on **Snowflake, Amazon Redshift, and BigQuery** processing **50TB+ monthly**, supporting 500+ business users with self-service analytics capabilities.
- Designed **Star and Snowflake schemas** optimizing query performance by **80%**, reducing executive report generation time from **10 minutes to 2 minutes** for C-suite stakeholders.
- Developed **100+ production ETL/ELT pipelines** using **Python, Apache Airflow, and dbt**, achieving **99.7% reliability** with automated data validation across **15M daily records**.
- Implemented **FinOps cost optimization** strategies including warehouse auto-suspend, query result caching, and workload management, saving **\$380K annually (35% reduction)** in cloud data warehouse spend.
- Created **data observability framework** with freshness SLAs, anomaly detection, and automated alerting, reducing mean time to detection (**MTTD**) from hours to **5 minutes** for critical incidents.
- Grew self-service reporting adoption from **20% to 75%** of total analytics consumption by delivering curated dimensional models enabling Tableau and Looker rollout.
- Resolved **200+ production incidents** through on-call rotations; established runbooks and durable fixes that reduced recurring incidents by **68%**.
- Mentored **4 junior engineers** on SQL optimization, dimensional modeling, and ETL best practices, cutting ramp-up time by **50%** through structured pairing and code review.

Data Engineer

United Techno — *Technology company; centralized data infrastructure and analytics pipelines.*

Jan 2014 – Dec 2016

- Built centralized **data warehouse** using **Data Vault 2.0** methodology, consolidating data from 12+ SaaS platforms and reducing report generation time by **70%** for business stakeholders.
- Processed large-scale datasets using **Hadoop ecosystem (HDFS, Hive)** supporting **5TB+ batch analytics** workloads and enabling historical trend analysis for product teams.
- Implemented **event-driven data pipelines** capturing **2M+ application events daily**, enabling real-time system monitoring and reducing incident detection time by **85%**.
- Automated infrastructure provisioning using **Terraform (IaC)**, reducing environment setup time from **3 days to 2 hours** and eliminating configuration drift across dev/staging/production.
- Optimized **PostgreSQL** and **MySQL** queries through indexing strategies and query rewriting, improving dashboard load times by **90%** (30 seconds → 3 seconds).
- Developed **25+ Power BI dashboards** tracking operational KPIs, providing executives with real-time business performance visibility.
- Added **data validation**, logging, and comprehensive monitoring checks to improve pipeline stability and reduce data quality issues by **60%**.
- Collaborated with cross-functional teams including engineers, analysts, and business stakeholders to deliver **production-ready datasets** meeting enterprise reporting requirements.

Key Projects

Enterprise-Scale Data Mesh with Federated Governance

- **Challenge:** Centralized data platform bottleneck causing **6+ week delays** in analytics delivery, limiting domain autonomy across 8 business units.
- **Action:** Architected enterprise **Data Mesh** on **Databricks**, **Delta Lake**, and **Amazon S3** with federated governance via **Unity Catalog** and **AWS Lake Formation**; implemented standardized data contracts, automated quality checks with **Great Expectations**, and self-service data product catalog.
- **Result:** Reduced data onboarding from 6 weeks to **48 hours**; enabled 8 domain teams to publish 50+ data products independently; achieved **99.8% data quality compliance**; maintained HIPAA audit readiness across all federated domains.

Zero-Downtime Data Platform Migration (Dual-Run Strategy)

- **Challenge:** Legacy warehouse hitting scalability limits with **\$2M+ migration risk** and business-critical analytics requiring 24/7 availability during transition.
- **Action:** Led dual-run migration to modern **lakehouse** using **Delta Lake** and **Apache Iceberg** across AWS/Azure; implemented parallel **Spark/Airflow** pipelines with automated 99.99% accuracy validation and phased cutover of **200+ Power BI/Tableau** dashboards.
- **Result:** Achieved **zero business downtime**; validated 100% data accuracy across **12 petabytes**; reduced annual cloud costs by **\$800K (32%)**; improved query performance by **4x** while maintaining all SLAs.

Real-Time Data Contracts & Schema Enforcement Platform

- **Challenge:** Breaking schema changes causing **15+ production incidents monthly**, eroding downstream analytics trust.
- **Action:** Built schema enforcement platform for streaming/CDC workloads using **Apache Kafka**, **Debezium**, and **Spark Structured Streaming** with CI/CD-enforced backward compatibility via **GitHub Actions**; integrated **Great Expectations** for real-time validation and **Prometheus/Grafana** alerting.
- **Result:** Eliminated **100% of breaking schema changes**; reduced data incidents by **92%** (15 → 1/month); cut mean time to detection from **4 hours to 30 seconds**; achieved **95% consumer confidence** rating.

Certifications

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| • Databricks Certified Data Engineer – Professional | 2023 |
| • AWS Certified Data Analytics – Specialty | 2022 |
| • Google Professional Data Engineer | 2021 |
| • Certified Data Management Professional (CDMP) | 2020 |

Education
